

1. Which members of the Circle class are encapsulated?

The data members (instance variables), especially those declared private (e.g., radius), are encapsulated.

2. What name must the constructor of a class have?

The constructor must have the exact same name as the class.

3. Explain the difference between private and public access modifiers.

private: Accessible only within the same class.

public: Accessible from any other class.

4. Is this valid? Explain.

```
Circle dot = new Circle(2);  
dot.radius = 5;
```

Invalid. If radius is private, it cannot be accessed directly. A setter method must be used.

5. Roo Class

```
public class Roo {  
    private int x;
```

```
    public Roo() {  
        x = 1;  
    }
```

```
    public void setX(int z) {  
        x = z;  
    }
```

```
    public int getX() {  
        return(x);  
    }
```

```
    public int calculate() {  
        x = x * factor();  
        return(x);  
    }
```

```
    private int factor() {  
        return(0.12);  
    }  
}
```

- a) Class name: Roo
- b) Data member: x
- c) Accessor method: getX()
- d) Modifier method: setX()
- e) Helper method: factor()
- f) Constructor: Roo()
- g) Number of methods: 5

6. Difference between a class and an object:

A class is a blueprint. An object is an instance of that class.

7. Band Festival OOP

- a) Class: Band
- b) Objects: TwoToos, EggRolls, Goop
- c) Data member: setList
- d) Methods: TuneUp(), PlayMusic(), TakeABow(), ReadSetList(), CreateSetList()

8. Team Class

- a) Objects: teamA, teamB, teamC
- b) Methods: playGame(), scorePoint(), train()
- c) Data members: teamName, score, numberOfPlayers

9. Moo Class

```
public class Moo {  
    private double y;  
    private static int x;  
    private static final z;  
}
```

- a) Constant: z
- b) Variables: y, x
- c) Instance members: y
- d) Class members: x, z

10. Instance vs Class Methods

Instance method: called using an object → obj.method()

Class method: called using class name → ClassName.method()

11. Overriding vs Overloading

Overriding: same method in subclass.

Overloading: same method name, different parameters.

12. Has-a Relationship

A class contains another object. Example: Customer has a BankAccount.

13. Code Reuse and Bugs

Reused code is already tested, so it reduces bugs.

14. Passing a Message

Calling a method on an object to perform an action.

True / False

15.

- a) False, State is described by variables.
- b) False Behavior is described by methods.
- c) True
- d) True
- e) False, Encapsulation hides variables.
- f) True
- g) True
- h) False, Accessors get values.
- i) False, Modifiers change values.
- j) True
- k) True
- l) True
- m) False, Class variables use static.
- n) False, Class methods use class name.
- o) False, Constants use static final.
- p) False, Object is the superclass of all classes.
- q) True
- r) False, OOP uses many objects.